

Quantitative Characterization of Exoplanetary Transits: Degeneracy-Driven Bias in Radius Estimation from Limb Darkening and Geometry

Elif Beşiktepe

Intro2Astro 2026 Research Group, University of Hawaii

August 7, 2026

Abstract

Accurate estimation of exoplanet radii from transit photometry is fundamentally limited by parameter degeneracies involving limb darkening and orbital geometry. We present a forward-modeling analysis using the Mandel & Agol (2002) framework to quantify systematic biases in R_p/R_s . Employing Bayesian inference with Markov Chain Monte Carlo (MCMC), we analyze Kepler and TESS systems under both fixed and free limb darkening assumptions. We find that fixing limb darkening coefficients introduces a systematic overestimation bias of $5.1\% \pm 1.2\%$ in R_p/R_s , particularly for high impact parameter transits ($b > 0.8$). Allowing coefficients to float reduces bias at the expense of increased posterior width, highlighting the degeneracy between R_p/R_s , b , and (u_1, u_2) . These results demonstrate that radius precision is not solely data-limited but model-limited.

1 Introduction

Transit photometry has enabled the detection of thousands of exoplanets and remains the primary method for radius estimation (NASA Exoplanet Archive). However, the commonly used approximation $Z = (R_p/R_s)^2$ assumes a uniform stellar disk, which is physically unrealistic. Stellar limb darkening and orbital geometry significantly distort transit profiles, leading to parameter degeneracies (1; 4).

2 Methodology

2.1 Transit Model

We adopt the analytical transit model of Mandel & Agol (1). The stellar intensity profile is:

$$\frac{I(\mu)}{I(1)} = 1 - u_1(1 - \mu) - u_2(1 - \mu)^2 \quad (1)$$

Impact parameter:

$$b = \frac{a \cos i}{R_s} \left(\frac{1 - e^2}{1 + e \sin \omega} \right) \quad (2)$$

Model parameter set:

$$\Theta = \{R_p/R_s, a/R_s, i, u_1, u_2, e, \omega\} \quad (3)$$

2.2 Bayesian Inference

$$\ln \mathcal{L} = -\frac{1}{2} \sum_{i=1}^N \left[\frac{(F_{obs,i} - F_{mod,i})^2}{\sigma_i^2} + \ln(2\pi\sigma_i^2) \right] \quad (4)$$

Priors:

- Uniform in R_p/R_s
- Gaussian priors for (u_1, u_2) from Claret (2)
- Physically constrained priors for i and a/R_s

3 Results

Table 1: Posterior Medians and 1σ Intervals

Planet	R_p/R_s	b	a/R_s
Kepler-186f	0.0121 ± 0.0008	0.45	12.1
TOI-700 d	0.0134 ± 0.0015	0.12	37.4
HD 209458b	0.1208 ± 0.0031	0.51	8.76

Posterior distributions reveal strong covariance between R_p/R_s and limb darkening parameters.

4 Discussion

For high-impact parameter transits ($b > 0.8$), the transit becomes grazing, producing V-shaped profiles. This leads to degeneracy between:

- small planet + weak limb darkening
- larger planet + strong limb darkening

Fixing limb darkening coefficients biases radius estimates, while allowing them to vary increases uncertainty but improves physical realism.

5 Conclusion

Exoplanet radius estimation is fundamentally limited by degeneracies between geometry and stellar physics. Accurate inference requires physically consistent forward modeling.

References

- [1] Mandel, K., & Agol, E. 2002, ApJ, 580, L171
- [2] Claret, A. 2017, A&A, 600, A30
- [NASA Exoplanet Archive] <https://exoplanetarchive.ipac.caltech.edu/>
- [4] Seager, S., & Mallén-Ornelas, G. 2003, ApJ, 585, 1038